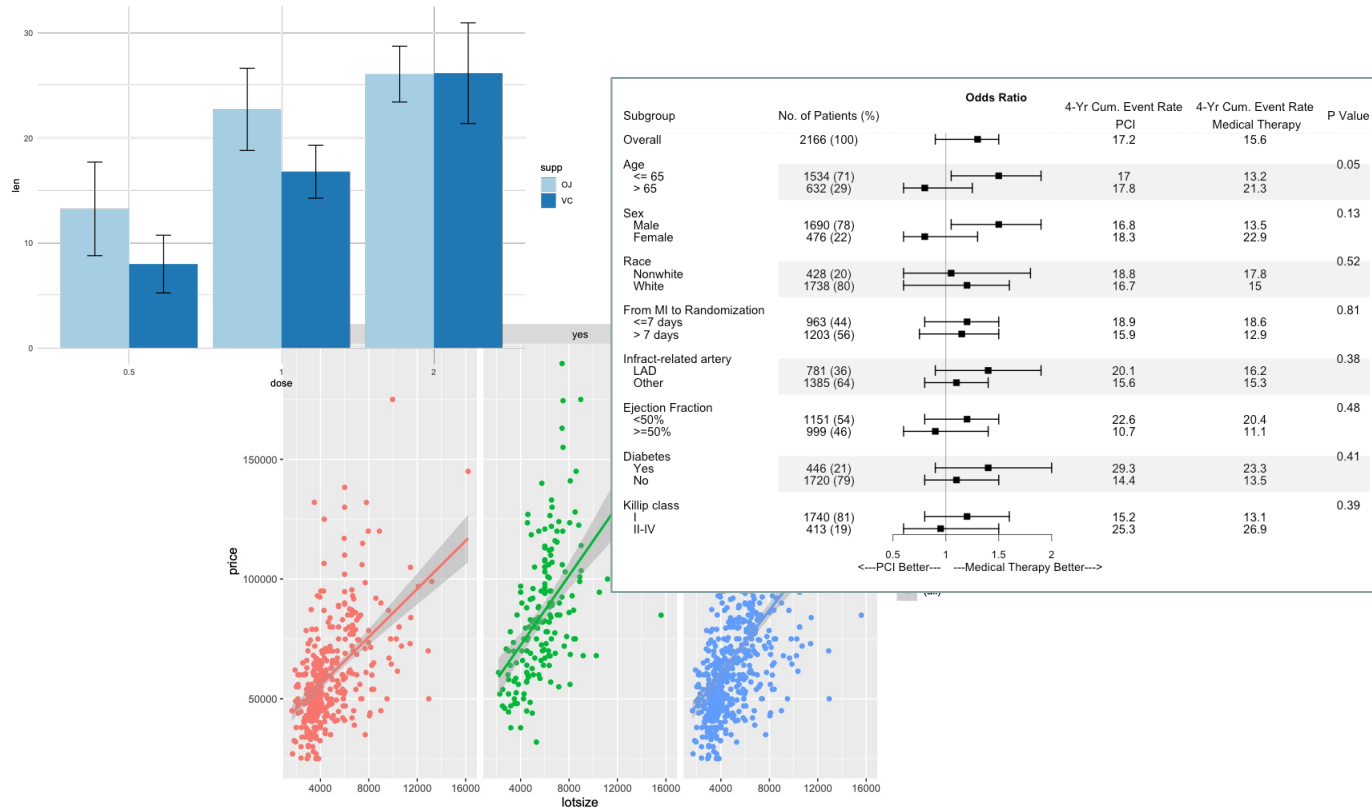


Introducción a





Manejo básico de datos

Fechas

Clase => “Date”

as.Date()

```
> date()
```

```
[1] "Sat May 14 21:01:58 2016"
```

- Internamente las fechas son representadas como el número de días después o antes de un punto conocido como “epoch” (January 1, 1970)

ID	edad	sexo	estado.civil	nivel.estudios	peso	altura	fumador	diabetes	cancer.mama	cancer.prostata	fdiag_cm	fdiag_cp	fdef
137	37	Mujer	Casado	Bajo	59.58221	150.7163	No	No	Si	<NA>	1977-05-05	<NA>	1977.05.14
174	85	Mujer	Soltero	Alto	59.95427	149.2075	No	Si	No	<NA>	<NA>	<NA>	<NA>
200	29	Hombre	Casado	Bajo	79.20674	168.9795	No	Si	Si	Si	2007-09-28	17.06.96	2007.12.13
23	13	Hombre	Divorciado	Alto	80.78347	171.1568	Si	Si	<NA>	Si	<NA>	12.08.06	2006.10.06
39	49	Hombre	Divorciado	Bajo	80.76036	170.5682	Si	Si	<NA>	No	<NA>	<NA>	<NA>
90	12	Hombre	Casado	Alto	79.83426	170.8565	No	No	<NA>	Si	<NA>	28.06.02	2002.08.21

```
> str(datos)
'data.frame': 200 obs. of 14 variables:
 $ ID      : num  137 174 200 23 39 90 40 115 72 27 ...
 $ edad    : num  37 85 29 13 49 12 85 31 39 70 ...
 $ sexo    : chr   "Mujer" "Mujer" "Hombre" "Hombre" ...
 $ estado.civil : chr   "Casado" "Soltero" "Casado" "Divorciado" ...
 $ nivel.estudios : chr   "Bajo" "Alto" "Bajo" "Alto" ...
 $ peso     : num   59.6 60 79.2 80.8 80.8 ...
 $ altura   : num   151 149 169 171 171 ...
 $ fumador   : chr   "No" "No" "No" "Si" ...
 $ diabetes  : chr   "No" "Si" "Si" "Si" ...
 $ cancer.mama : chr   "Si" "No" "Si" NA ...
 $ cancer.prostata: chr   NA NA "Si" "Si" ...
 $ fdiag_cm  : chr   "1977-05-05" NA "2007-09-28" NA ...
 $ fdiag_cp  : chr   NA NA "17.06.96" "12.08.06" ...
 $ fdef      : chr   "1977.05.14" NA "2007.12.13" "2006.10.06" ...
```

ID	fdiag_cm	fdiag_cp	fdef
137	1977-05-05	<NA>	1977.05.14
174	<NA>	<NA>	<NA>
200	2007-09-28	17.06.96	2007.12.13
23	<NA>	12.08.06	2006.10.06
39	<NA>	<NA>	<NA>
90	<NA>	28.06.02	2002.08.21
40	<NA>	29.08.02	2002.10.07
115	1980-01-01	<NA>	1980.03.24
72	1996-10-12	<NA>	1997.01.08
27	<NA>	23.04.01	2001.05.08

```
> class(datos$fdiag_cm)
```

```
[1] "character"
```

```
> fechas.CM <- as.Date(datos$"fdiag_cm")
```

```
> class(fechas.CM)
```

```
[1] "Date"
```

```
> fechas.CM[1:6]
```

```
[1] "1977-05-05" NA      "2007-09-28" NA      NA      NA
```

ID	fdiag_cm	fdiag_cp	fdef
137	1977-05-05	<NA>	1977.05.14
174	<NA>	<NA>	<NA>
200	2007-09-28	17.06.96	2007.12.13
23	<NA>	12.08.06	2006.10.06
39	<NA>	<NA>	<NA>
90	<NA>	28.06.02	2002.08.21
40	<NA>	29.08.02	2002.10.07
115	1980-01-01	<NA>	1980.03.24
72	1996-10-12	<NA>	1997.01.08
27	<NA>	23.04.01	2001.05.08

```
> class(datos$fdiag_cp)
```

```
[1] "character"
```

```
> fechas.CP <- as.Date(datos$"fdiag_cp", format="%d.%m.%y")
```

```
> class(fechas.CP)
```

```
[1] "Date"
```

```
> fechas.CP[1:6]
```

```
[1] NA      NA      "1996-06-17" "2006-08-12" NA      "2002-06-28"
```

ID	fdiag_cm	fdiag_cp	fdef
137	1977-05-05	<NA>	1977.05.14
174	<NA>	<NA>	<NA>
200	2007-09-28	17.06.96	2007.12.13
23	<NA>	12.08.06	2006.10.06
39	<NA>	<NA>	<NA>
90	<NA>	28.06.02	2002.08.21
40	<NA>	29.08.02	2002.10.07
115	1980-01-01	<NA>	1980.03.24
72	1996-10-12	<NA>	1997.01.08
27	<NA>	23.04.01	2001.05.08

```
> class(datos$fdef)
```

```
[1] "character"
```

```
➤ fechas.DF <- as.Date(datos$"fdef",format="%Y.%m.%d")
```

```
> class(fechas.DF)
```

```
[1] "Date"
```

```
> fechas.DF[1:6]
```

```
[1] "1977-05-14" NA "2007-12-13" "2006-10-06" NA "2002-08-21"
```

Formats Dates

Conversion specification	Description	Example
%a	Abbreviated weekday	Sun, Thu
%A	Full weekday	Sunday, Thursday
%b or %h	Abbreviated month	May, Jul
%B	Full month	May, July
%d	Day of the month 01-31	27, 07
%j	Day of the year 001-366	148, 188
%m	Month 01-12	05, 07
%U	Week 01-53 with Sunday as first day of the week	22, 27
%w	Weekday 0-6 Sunday is 0	0, 4
%W	Week 00-53 with Monday as first day of the week	21, 27
%x	Date, locale-specific	
%y	Year without century 00-99	84, 05
%Y	Year with century on input: 00 to 68 prefixed by 20 69 to 99 prefixed by 19	1984, 2005
%C	Century	19, 20
%D	Date formatted %m/%d/%y	05/27/84, 07/07/05
%u	Weekday 1-7 Monday is 1	7, 4
%n	Newline on output or Arbitrary whitespace on input	
%t	Tab on output or Arbitrary whitespace on input	

?strptime

```
> datos$"fechas.CM"<-fechas.CM  
> datos$"fechas.CP"<-fechas.CP  
> datos$"fechas.DF"<-fechas.DF
```

```
> head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF")])
```

ID	fechas.CM	fechas.CP	fechas.DF
137	1977-05-05	<NA>	1977-05-14
174	<NA>	<NA>	<NA>
200	2007-09-28	1996-06-17	2007-12-13
23	<NA>	2006-08-12	2006-10-06
39	<NA>	<NA>	<NA>
90	<NA>	2002-06-28	2002-08-21

```
> datos$"dias"<-c(datos$"fechas.DF" - datos$"fechas.CM")
```

```
> class(datos$"dias")  
[1] "difftime"
```

```
head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF","dias")])
```

ID	fechas.CM	fechas.CP	fechas.DF	dias
137	1977-05-05	<NA>	1977-05-14	9 days
174	<NA>	<NA>	<NA>	NA days
200	2007-09-28	1996-06-17	2007-12-13	76 days
23	<NA>	2006-08-12	2006-10-06	NA days
39	<NA>	<NA>	<NA>	NA days
90	<NA>	2002-06-28	2002-08-21	NA days

```
> datos$"dias2"<-difftime(datos$"fechas.DF", datos$"fechas.CM",units="days")
```

```
> class(datos$"dias2")
```

```
[1] "difftime"
```

```
> head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF","dias","dias2")])
```

ID	fechas.CM	fechas.CP	fechas.DF	dias	dias2
137	1977-05-05	<NA>	1977-05-14	9 days	9 days
174	<NA>	<NA>	<NA>	NA days	NA days
200	2007-09-28	1996-06-17	2007-12-13	76 days	76 days
23	<NA>	2006-08-12	2006-10-06	NA days	NA days
39	<NA>	<NA>	<NA>	NA days	NA days
90	<NA>	2002-06-28	2002-08-21	NA days	NA days

```
> datos$"weeks"<-difftime(datos$"fechas.DF", datos$"fechas.CM",units="weeks")
```

```
> class(datos$"weeks")  
[1] "difftime"
```

```
> head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF","dias","dias2","weeks")  
])
```

ID	fechas.CM	fechas.CP	fechas.DF	dias	dias2	weeks
137	1977-05-05	<NA>	1977-05-14	9 days	9 days	1.285714 weeks
174	<NA>	<NA>	<NA>	NA days	NA days	NA weeks
200	2007-09-28	1996-06-17	2007-12-13	76 days	76 days	10.857143 weeks
23	<NA>	2006-08-12	2006-10-06	NA days	NA days	NA weeks
39	<NA>	<NA>	<NA>	NA days	NA days	NA weeks
90	<NA>	2002-06-28	2002-08-21	NA days	NA days	NA weeks

?difftime

```
> datos$"years"<-difftime(datos$"fechas.DF", datos$"fechas.CM",units="days") / 365.25
> class(datos$"years")
[1] "difftime"
> head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF","dias","weeks","years")])
```

ID	fechas.CM	fechas.CP	fechas.DF	dias	weeks	years
137	1977-05-05	<NA>	1977-05-14	9 days	1.285714 weeks	0.02464066 days
174	<NA>	<NA>	<NA>	NA days	NA weeks	NA days
200	2007-09-28	1996-06-17	2007-12-13	76 days	10.857143 weeks	0.20807666 days
23	<NA>	2006-08-12	2006-10-06	NA days	NA weeks	NA days
39	<NA>	<NA>	<NA>	NA days	NA weeks	NA days
90	<NA>	2002-06-28	2002-08-21	NA days	NA weeks	NA days

```
> attr(datos$"years","units")="years"
> head(datos[,c("ID","fechas.CM","fechas.CP","fechas.DF","dias","weeks","years")])
```

ID	fechas.CM	fechas.CP	fechas.DF	dias	weeks	years
137	1977-05-05	<NA>	1977-05-14	9 days	1.285714 weeks	0.02464066 years
174	<NA>	<NA>	<NA>	NA days	NA weeks	NA years
200	2007-09-28	1996-06-17	2007-12-13	76 days	10.857143 weeks	0.20807666 years
23	<NA>	2006-08-12	2006-10-06	NA days	NA weeks	NA years
39	<NA>	<NA>	<NA>	NA days	NA weeks	NA years
90	<NA>	2002-06-28	2002-08-21	NA days	NA weeks	NA years

Otros ejemplos:

```
> dates <- c("01ene1960", "02ene1960", "31mar1960", "30jul1960")  
> as.Date(x=dates, format="%d%b%Y")
```

```
> dates <- c("02/27/92", "02/27/92", "01/14/92", "02/28/92", "02/01/92")  
> as.Date(x=dates, format="%m/%d/%y")
```

Recodificar una variable fecha a una de caracteres con un formato:

```
> datos$DF_new <- as.character(format(datos$"fechas.DF",format="%m/%Y"))  
  
> head(datos[,c("ID","fechas.DF","DF_new")])
```

ID	fechas.DF	DF_new
137	1977-05-14	05/1977
174	<NA>	<NA>
200	2007-12-13	12/2007
23	2006-10-06	10/2006
39	<NA>	<NA>
90	2002-08-21	08/2002

Otras operaciones con fechas

```
> datos$"fecha.DF_rev"<-datos$"fechas.DF" - 5
```

```
> head(datos[,c("ID","fechas.DF","fecha.DF_rev")])
```

ID	fechas.DF	fecha.DF_rev
137	1977-05-14	1977-05-09
174	<NA>	<NA>
200	2007-12-13	2007-12-08
23	2006-10-06	2006-10-01
39	<NA>	<NA>
90	2002-08-21	2002-08-16

[ejercicios.fechas.pdf](#)